NOT MEASUREMENT SENSITIVE

MIL-T-46765D(AT)
07 February 1992
SUPERSEDING
MIL-T-46765C(AT)
16 January 1981

MILITARY SPECIFICATION

TRUCKS, TANK: SEWAGE AND WASTE, DUMP TYPE, 4550 AND 7570 LITER (1200 AND 2000 GALLON), MODIFIED COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel and gasoline engine driven trucks equipped with a hydraulic lift dump tank, an auxiliary engine driven centrifugal pump, and associated hoses and equipment for loading and unloading the dump tank with sewage and waste materials. The vehicle is capable of cleaning sewers, septic tanks, catch basins, manholes, and sludge pits. The trucks have minimum gross vehicle weights (GVW) of 11 350 kilograms (kg) (25,000 pounds) in a four wheel, two wheel drive (4x2), 4550 liter (L) (1200 gallon) size; and 15 650 kg (34,500 pounds) in a six wheel, four wheel drive (6x4), 7570 L (2000 gallon) size; for normal rating conditions. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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1.2 Classification. The truck shall be one of the following sizes, as specified (see 6.2):

Size 1200	- 4550 L (1200 gallon) capacity tank; 11350 kg (25,000 pounds) GVW, 4x2 chassis.
Size 2000	- 7570 L (2000 gallon) capacity tank; 15 650 kg (34,500 pounds) GVW, 6x4 chassis.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

* 2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting, and Starting.
*	VV-F-800	- Fuel, Oil, Diesel.

MILITARY

*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

*	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.
	FED-STD-595	- Colors.

MILITARY

*	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
	MS 51118	- Pintle Assembly, Towing – 40,000 Lbs. Capacity, Manual Release.
	MS 51317	- Light, Warning, Vehicular, Rotating, D.C.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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* 2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

* DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

* (Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment -Medium and Heavy Trucks .

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 4985	- Standard Specification for Low Silicate Ethylene Glycol Base Engine Coolant for Heavy Duty Engines Requiring an Initial Charge of Supplemental Coolant Additive.

(Application for copies of ASTM publications should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

* NATIONAL TRUCK EQUIPMENT ASSOCIATION (NTEA)
Dump Body Hoist Chart.

(Application for copies of NTEA publications should be addressed to the National Truck Equipment Association, 38705 Seven Mile Road, Suite 345, Livonia, MI 48152.)

SAE, INC.
SAE Standards and Recommended Practices

	J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers –Trucks, Truck-Tractors, and Trailers (DoD Adopted).
	J537	- Storage Batteries (DoD Adopted).
*	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J598	- Sealed Lighting Units for Construction and Industrial Machinery.
	J682	- Rear Wheel Splash and Stone Throw Protection (DOD Adopted).
	J688	- Truck Ability Prediction Procedure .
*	J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs (DOD Adopted).
	J816	- Engine Test Code - Spark Ignition and Diesel.
	J844	- Nonmetallic Air Brake System Tubing (DOD Adopted).
*	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

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* (Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government Standards and other publications are normally available from the Organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.6, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.6, no removal, substitution or alternation of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. Treatment, painting, marking and data plates shall be in accordance with MIL-STD-1223. The exterior color and the identification markings shall be in accordance with the requirements of MIL-STD-1223 for the military service identified by the procuring activity (see 6.2). When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.3 Drain plugs. Drain plugs installed in manual transmissions and rear axles shall be of the permanent magnet type.

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* 3.1.1.4 Towing devices. Towing devices consisting of two hooks, loops, eyes or pins or the chassis manufacturer's standard single center mounted eye or pin shall be mounted on the front of the vehicle. When specified (see 6.2), in addition, not less than two hooks, loops or pins for towing the vehicle shall be furnished on the rear of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.5 Wheel splash and stone throw protection. The truck shall have rigid splash shields ahead of the rear wheels and rubber mud flaps to the rear of the rear wheels. All tilt cabs shall have rubber mud flaps to the rear of the front wheels. A metal strip not less than 3.2 millimeters (mm) (0.125 inch) thick and not less than 25 mm (1 inch) wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.250 inch) wide by 2.4 mm (0.094 inch) thick shall be furnished at each bolt. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

3.1.1.6 Trailer towing package. When specified (see 6.2), a trailer towing package shall be furnished. The trailer towing package shall consist of a pintle, safety chain attachment devices, a lighting receptacle, a trailer brake control system, and associated reinforcement and wiring, and shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS 51118. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 1815 kg (4,000 pounds) and a horizontal drawbar load of not less than 178 kilonewtons (kN) (40,000 pounds) directly to the chassis rails. The rearmost portion of the pintle shall be forward, but not more than 100 mm (four inches) forward, of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The attachment devices shall be capable of accommodating a standard grab hook [116 mm (4-9/16 inches) wide, 30 mm (1-3/16 inches) thick, 19.8 mm (23/32-inch) throat width)] for a 16 mm (5/8-inch) chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color-coded as specified herein, or number codes, shall be mounted in a readily accessible location near the pintle. The trailer brake control system shall conform to 3.4.11.2.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

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3.2.2 Air pollution control. The vehicle shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California Regulations governing Air Pollution Control in effect on the date of manufacture.

* 3.2.3 Sound level. The cab interior sound level shall not exceed 84 db(A) when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to the EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. Curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; the body; and a full complement of fuel, lubricants, and coolant.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 80 kg (175 pounds)), and a payload to provide not less than the specified GVW.

3.2.6 Weight distribution. The distribution of GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be submitted to the engineering office of the procuring activity. The minimum GVW rating shall conform to table I for the specified size of vehicle.

TABLE I. GVW minimum rating.

Vehicle size	Rating	
	kg	(pounds)
1200	11 350	(25,000)
2000	15 650	(34,500)

* 3.2.8 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

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* 3.2.9 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part of assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with truck loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend continuous grades specified in table II at 80 kilometers per hour (km/h) (50 miles per hour (mph)). Gradeability requirements shall be met with the main transmission in direct drive, and when a multi-speed axle is furnished, with the axle in high speed range. Gradeabilitysha11 be verified with calculations in accordance with SAE J688 (see 6.3).

* TABLE II. Gradeability.

Vehicle size	Percent of grade
1200	2.3
2000	1.5

3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed, and shall provide not more than the vehicle speed (mph) specified in table III for the corresponding number of available forward speeds.

TABLE III. Low speed requirement.

	Number of forward speeds	Maximum low speed km/h (mph)
	5	7.2 (4.5)
*	7 or 8	5.6 (3.5)
*	9 or 10	4.8 (3.0)

3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 93 km/h (58 mph). Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (km/h)} \quad \frac{\text{Governed speed (rpm)} \times 1.609}{\text{Total gear reduction x tire factor (see 6.3)}}$$

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3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine. Engine net power used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be supplied with a liquid cooled, internal combustion, four-stroke cycle gasoline engine. The engine furnished shall provide the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of continuous, warranted operation on unleaded fuel. Engine net power figures used in performance prediction calculations shall be determined in accordance with SAE J1349.

* 3.4.1.3 Oil filter. A full flow or combination full flow and bypass oil filter with replaceable element shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to specified GVW and operated at an altitude of 3050 meters (m) (10,000 feet) above sea level or in an ambient air temperature of not less than 52° Celsius (°C) (125 degrees Fahrenheit (°F)). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than 1.89 liters (L) (two-quart) capacity. On tilt cab models, a radiator servicing access door shall be provided to allow verification of the coolant level. For cooling system servicing, see 3.4.25.1.

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* 3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

* 3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.9 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater; an engine oil heater; and fuel warmers (diesel engine driven vehicles only) shall be provided. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding vehicle. The receptacle shall be located on the streetside of the vehicle, as near the cab door as possible. A three-wire connecting cable, not less than 7600 mm (25 feet) long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature (down to -51°C (-60°F)) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters and fuel warmers shall be furnished as follows:

* (a) A coolant heater, 1000-watt minimum rating, shall be installed in the engine block or lower, coolant, inlet hose. An engine thermostat with an operating range of 77°C to 90°C (170°F to 195°F) shall be installed;
* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:

    * (1) Not less than 16 W/L (15 watts per quart) heating capacity
    * (2) Not more than 2.8 watts per square centimeter (W/cm^2) (18 watts per square inch) heating capacity
    (3) Thermal balance design or thermostat control providing for uninterrupted operation
    (4) Provision for mounting below minimum service oil level.

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* (c)An in-line fuel warmer or preheater unit shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The unit shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°C (-40°F) to an outlet temperature of -13°C (+9°F), with a fuel flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit. The unit shall not cause heating of the fuel above 27°C (80°F) under any possible condition.

* (d) An in-tank fuel warmer or preheater unit shall be provided on diesel engine driven vehicles. The unit shall use engine coolant to transfer heat to the fuel in one fuel tank. The warmer shall not cause heating of any fuel above 27°C (80°F) under any possible condition, shall not disable or cause elimination of the fuel gage sending unit and shall not violate 3.2.1 or any Federal Motor Carrier Safety Regulation. A coolant shut off valve shall be included. The units required by 3.4.1.9 (c) and (d) may be combined.

* 3.4.1.10 Vernier throttle control. When specified (see 6.2), a manual vernier controlled throttle with quick release shall be furnished, except on electronically controlled engines.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with a 12-volt dc lighting system shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or electric grid heater. If an ether system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 355 milliliters (mL) (12 fluid ounces) shall be furnished. A reservoir need not be furnished. When specified (see 6.2), the starting motor shall be equipped with a thermostat controlled, automatic resetting circuit breaker that will protect the motor from overcrank heat damage. The circuit breaker shall not limit cranking ability for a startable engine.

* 3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished.

* 3.4.2.3 Alternator. Diesel engine driven vehicles shall have a minimum 65-ampere alternator. The alternator output with the engine at engine idle speed shall be not less than 45 amperes. Unless otherwise specified (see 6.2) gasoline engine driven vehicles shall have an

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alternator of not less than 85 amperes rated capacity, and which provides not less than 45 amperes dc output at normal engine idle speed.

* 3.4.2.4 Lighting. All vehicle lights, reflectors, and wiring shall conform with Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surfaces or rub rails (unless recessed and fully protected) or mounted on vehicle bumpers.

* 3.4.2.5 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Turn signal units shall be installed in accordance with SAE 3588. Turn signal operating units shall have a visible flash indicator.

3.4.2.6 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating and total cold cranking ampere rating at -18°C (0°F), both measured in accordance with SAE J537, shall be not less than specified in table IV. Batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

* TABLE IV. Batteries.

Engine Type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	480	1,875
* Gasoline	100	450

* 3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) shall be not less than 189 L (50 gallons) total capacity (163 L (43 gallons for vehicles destined for California)). When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure an equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve connecting either tank to engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage; or an equalizing pump shall be used to maintain the same fuel level in both tanks.

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* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. Vertical exhaust mufflers, capable of being reached easily by personnel entering or leaving either side of the cab, shall be furnished with a heat shield. Vertical exhaust systems shall be provided with a hinged rain cap.

3.4.5 Transmission. Unless otherwise specified (see 3.4.5.2), a manually shifted transmission shall be provided on the vehicle. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The vehicle shall be provided with not less than the number of forward speeds specified in table V. When more than five forward speeds are required, a multispeed transmission, auxiliary transmission or two-speed axles shall be provided. Gear ratios in transmissions and axles shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide required vehicle performance (see 3.3.1). The transmission shall provide for maximum ease of shifting in all speeds. The transmission shall be provided with SAE J704 power takeoff openings at right and left sides.

Table V. Transmission requirements, minimum number of forward speeds.

Vehicle size	Minimum number of forward speeds
1200	5
* 2000	9

* 3.4.5.1 Clutch. The clutch shall be the largest capacity clutch offered for the size vehicle and engine furnished with clutch torque capacity exceeding maximum delivered engine torque. The clutch lining shall be asbestos free. The clutch on size 2000 vehicle shall have spring dampening and a greaseable bearing.

* 3.4.5.2 Automatic transmission. When specified (see 6.2), an automatic transmission shall be furnished. The transmission shall include a hydraulic torque convertor and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four forward gear ratios without movement of the selector. The transmission shall have a power takeoff opening.

3.4.5.3 Power takeoff. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

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3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The vehicle shall have a heavy duty main frame or frame reinforcements. The heavy duty frame shall have frame rails of greater section modulus than the manufacturer's standard for the size vehicle furnished and shall provide structural strength at least equivalent to the reinforced frame specified herein. Reinforcements shall extend at least from the rear of the front suspension, rear hanger bracket to the front of the rear spring, front hanger bracket for size 1200 vehicles. Reinforcements for size 2000 vehicles shall extend at least from the rear of the front suspension, rear hanger bracket to the bogie trunnion mounting bracket. Reinforcements shall provide sufficient structural strength in the chassis frame through increased resisting bending moment to at least equal the loads imposed, with the dump body loaded to provide specified GVW.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system, with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to specified GVW. When suspension is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double- acting shock absorbers at the front wheels.

* 3.4.9 Ax1es. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. The wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Two-speed axle. When specified (see 6.2), or when furnished in accordance with 3.4.5, a two-speed axle shall be furnished, equipped with electric, vacuum, or air shift and provided with ratios which will permit proper gear splitting. The gear ratio shall provide the performance specified in 3.3.1.

3.4.9.2 Rear bogie. The rear bogie on size 2000 vehicles shall be of the four-wheel type, complete with axles, springs, torque rods, and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles, and a manually or automatically controlled lockout assuring equal power to each rear axle. The manual lockout control used shall be located in the truck cab. Axle gear ratios shall provide performance specified in 3.3.1.

3.4.9.3 Traction control. When specified (see 6.2), traction control shall be furnished on one rear axle. The traction control shall actuate automatically to insure that power is transmitted to

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the wheels having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires, and tubes. Unless otherwise specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations, for the type and size of tires furnished. Tire, rim, and disc wheel sizes shall be the same for all wheels on each vehicle. Multi-piece rims shall not be furnished. When specified (see 6.2), the vehicle shall be equipped with wide base type tires and wheels for the front and rear axle in lieu of single front and dual rears. Wide base wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), 10 stud disc type wheels shall be furnished.

* 3.4.10.1 Tires. Tires shall be steel belted radial or, when specified (see 6.2), bias ply. Unless otherwise specified, low profile or standard profile tires may be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall have highway tread. Tires shall be of the tubeless type. Tires shall be of a rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the vehicle loaded to specified GVW. Tires shall ccnform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.4.10.2 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable removal or mounting of the spare wheel assembly using only tools specified in 3.4.16.1. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

* 3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42, and 393.45 through 393.52. Brake lining shall be of nonasbestos material.

* 3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head-type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 205 L/m (7.25 cubic feet per minute (cfm)) for size 1200 trucks and 340 L/m (12 cfm) for size 2000 trucks
(b) Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank

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(c) Foot control, suspended or treadle type
* (d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low-air-pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
* (i) Automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes.

* 3.4.11.1.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.2 Trailer brake control system. When a trailer towing package is specified (see 3.1.1.6), a trailer brake control system shall be furnished. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control
(c) Independent hand control for trailer brakes
(d) Prime mover protection valve with dash control and automatic breakaway feature
* (e) Trailer stoplight control operable with foot brake and with hand control for trailer brakes
(f) Two SAE J844 coiled air hoses, not less than 2800 mm (110 inches) long when fully extended, with SAE J318 gladhand couplers on both ends of hoses. The hoses shall be packaged and stowed in the vehicle for shipment;
* (g) Air connectors for trailers with SAE J318 gladhand couplers mounted at the rear of the vehicle located to prevent interference with a trailer
(h) Dummy gladhand couplers with chains
* (i) Prime mover only parking brake value to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.3 Brake control from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains.

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The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab type. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional full width model cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism; or a vehicle with a butterfly hood and with bolt on fenders capable of being removed or swinging forward when unbolted; shall be furnished. Tilting shall not interfere with present or future installations of additional equipment, such as a snowplow or front mounted winch. When specified (see 6.2), a tilt type cab with tilting and locking mechanism shall be furnished. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 610 mm (24 inches).

* 3.4.12.1 Cab features. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage with both doors equipped with external, key-operated locks. Drip rails shall be installed above the cab doors. Safety grips or grab handles shall be provided on each side of the cab to assist personnel in entering and leaving the cab. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be furnished. For tilt type cabs, provisions to facilitate cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield.

* 3.4.12.2 Cab interior. Unless otherwise specified, the cab shall have upholstered, full-width, adjustable seat and back or individual, adjustable driver's seat and individual passenger seat. When specified (see 6.2), an individual adjustable driver's seat and an individual passenger seat shall be provided. When specified (see 6.2), the driver's seat shall be the manufacturer's standard air ride suspension type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be provided. Interior lighting shall be provided. Two pairs of seat belts shall be installed. Outboard seats shall have combination pelvic and upper torso restraint seat belts.

3.4.13 Steering. Power steering shall be furnished

* 3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by either air or electric motor(s). When specified (see 6.2), and if electric motor wipers are furnished, intermittent wipers shall be furnished.

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3.4.15 Bumpers. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided. Rear end protection in accordance with Federal Motor Carrier Safety Regulation 393.86 shall be furnished.

3.4.16 Tool stowage. Stowage space of sufficient size to accommodate vehicle jack, hand tools, antiskid chains (for outside tires on duals only), and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of the cab, it shall be waterproof and shall provide for locking with a padlock.

* 3.4.16.1 Tools. When specified (see 6.2), each vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under an axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of wheel and tire assembly.

3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes and discharge outlets to floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in driving position.

* 3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator, or voltmeter
(c) Fuel gage
* (d) Oil pressure gage or red indicator warning light
* (e) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Ash receptacle

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(i) Dual sunvisors
(j) Driver's compartment ventilator other than window
(k) Tachometer (diesel engine driven vehicles only)

When specified (see 6.2), an ammeter or voltmeter gage, oil pressure gage, and coolant temperature gage shall be provided in lieu of the indicators and buzzers specified above. When specified (see 6.2), an automatic engine shutdown system also shall be furnished. The system shall include an engine coolant temperature, engine coolant level and engine oil pressure red indicator warning light and alarm buzzer. The warning light and buzzer actuation shall precede engine shutdown. The system shall permit engine restart and run for approximately 30 seconds follow automatic shutdown. Also, engine oil pressure and engine coolant temperature gages shall be provided with the automatic engine shutdown system.

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on both sides of the cab. The mirrors shall have flat and convex areas. The flat portion shall have not less than 320 square centimeters (cm^2) (50 square inches) of reflective area. The convex portion shall have not less than 155 cm^2 (24 square inches) of reflective area and a radius of curvature of not less than 510 mm (20 inches). The convex portion shall be attached to the lower mirror supporting arm and shall not interfere with use of the flat mirror. The mirrors shall have not less than two supporting arms. When specified (see 6.2), the curbside flat mirror shall be of the motorized type, with remote control. The mirror motor shall provide not less than 60 degrees horizontal rotational viewing range. When specified (see 6.2), the flat mirrors shall be electrically heated. Mirror remote and heating controls shall be within reach of the seated driver.

3.4.21 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see .6.2), in addition, an air operated horn shall be provided.

3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel.

3.4.23 Rotating warning light. An amber, rotating, vehicular warning light conforming to MS 51317-4 shall be provided. The rotating light shall be mounted with reinforcement at point of attachment on vehicle centerline for 360 degree visibility. The warning light shall be operated by means of a separate switch with red indicator light located on instrument panel and marked to indicate function of switch.

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* 3.4.24 AM/FM radio. When specified (see 6.2), the manufacturer's standard AM/FM radio shall be provided.

* 3.4.25 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustment of engine; adjustment of brake system; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system in accordance with 3.4.25.1; and servicing of windshield washer reservoir with water and appropriate additives.

* 3.4.25.1 Engine coolant. The engine coolant shall be a solution of ethylene glycol antifreeze and water or propylene glycol antifreeze and water, in equal parts of antifreeze and water by volume. Ethylene glycol antifreeze shall conform to ASTM D 4985 with not more than 250 parts per million silicates. Propylene glycol antifreeze shall be in the specific formulation approved by the engine and cooling system manufacturers. When specified (see 6.2), the percentage of antifreeze in the cooling system shall be increased to provide protection against freezing down to -54°C (-65°F).

3.5 Body. The truck shall be equipped with a hydraulic lift operated dump type rectangular tank, an auxiliary engine driven centrifugal pump, and associated hoses and equipment suitable for loading, transporting and unloading heavy sewage and waste materials. The cab-to-axle or cab-to-trunnion dimension shall be not less than 3050 mm (120 inches).

3.5.1 Tank body. The tank body shall be equipped with a watertight rear tailgate hinged at the top, a top deck with hinged covers fully enclosing the tank, a hatch, baffles, a sewage and waste inlet, a level indicator, a drain, a hydrant water inlet, and a hydraulically operated dump body type hoist. The tank body shall be watertight to a water level capacity of 4.6 m^3 (6 cubic yards) or 4550 L (1200 gallons) for size 1200 trucks and 7.6 m^3 (10 cubic yards) or 7570 L (2000 gallons) for size 2000 trucks.

3.5.2 Tank body construction. For size-1200 trucks, inside length shall not be less than 2740 mm (108 inches) and inside width shal1 be not less than 1980 mm (78 inches). For size 2000 trucks, inside length shall be not less than 3050 mm (120 inches) and inside width shall be not less than 2130 mm (84 inches). Body floor, sides, and front head shall be of not less than 10 gage (3.416 mm) (0.1345 inch) steel. When the body floor in constructed in two or more pieces, an automatic seam weld having full penetration is acceptable without reinforcement. When the seam is hand welded, the underside of joint shall be reinforced with 75 mm (3-inch) wide, 8 gage (4.176 mm) (0.1644 inch) butt plates, welded between crossmembers. Full-length,

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channel formed rub rail of minimum width to cover rear dual tire treads shall be provided. Front head sheet shall be formed or reinforced for rigidity. Front head and tailgate shall be the same height as the sides. The interior of body shell and side reinforcements shall be welded with continuous welds. The rub rail shall be attached using continuous welds, or with intermittent welds for not less than 50 percent of the edge length of contact edges. The body interior shell shall be watertight and of smooth skin construction.

3.5.2.1 Tailgate. The tailgate panel shall be not less than 10 gage (3.416 mm) (0.1345 inch) steel. When tailgate is closed, the tailgate shall be watertight. The tailgate shall have heavy duty hardware and a heavy duty tailgate latch operable by a control at the front curbside corner of the tank body. The tailgate shall be reinforced to prevent deformation under load.

3.5.2.2 Body understructure. Body longitudinal sills, each having a minimum section modulus equivalent to that provided by a 125 mm (5 inches) by 14.9 kg/m (10 pounds per foot) I-beam, shall be provided to support hoist load. Not less than eight crossmembers, each having a minimum section modulus equivalent to that provided by a 100 mm (4 inches) by 8 kg/m (5.4 pounds per foot) channel shall be provided to support an evenly distributed load. Crossmembers shall be welded to the body shell with not less than 100 mm (4-inch) lengths of weld, front and rear of both ends of each crossmember, and with staggered intermittent welds on not less than 300 mm (12-inch) centers. Contact edges of crossmembers with longitudinal sills, and contact edges of welded reinforcements shall be welded for not less than 50 percent of the edge length.

3.5.2.3 Baffles. Three baffles shall be furnished. The baffles shall be minimum 10 gage (3.416 mm) (0.1345 inch) steel. The baffles shall be installed vertically and shall be hinged at the top so that the baffles swing back when the tank body is elevated for discharging the tank contents. Stops shall be provided to prevent the baffles from swinging forward. The baffles shall be the height and width of the interior of the body tank, and have a minimum clearance on the bottom. The front baffle shall have staggered, 19 mm (.750-inch) diameter holes with a center to center distance of approximately one inch. The open area shall be 51 percent of the baffle area. Hydraulically operated baffle agitator shall be furnished to clear the sewage waste from the front baffle and permit the flow of clear water into the front compartment. The two rear baffles shall be solid. Each of the two rear baffles shall have a notch on the top edge of sufficient size to allow flow of waste material from one compartment to another. One baffle shall have the notch on the street side and the other baffle shall have the notch on the curb side. The purpose of the notches is for overflow of water from one compartment to the next compartment. The forward baffle shall strain out the sewage to a liquid state. The rear baffle shall be approximately 1220 mm (48 inches) from the tailgate. The two forward baffles shall be located so that the three forward compartments are of equal capacity.

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3.5.2.4 Body tank cover (deck). A deck shall be installed completely around the inside perimeter of the tank. The width of the rear deck shall be approximately 529 mm (20.75 inches). The width of the two side decks measured from the outside edge shall be approximately 523 mm (20.5 inches). The width of the front deck shall be approximately 560 mm (22 inches). The deck shall be arc welded to the body shell and shall be watertight. The remaining open area shall be enclosed with two, 10 gage (3.416 mm) (0.1345 inch) steel hinged covers with gaskets, positioned so that the covers may be opened for inspection of the interior. A suitable locking means shall be provided to prevent covers from opening while the tank is in the hoisting or traveling position.

3.5.2.5 Hatch. A hatch with cover shall be provided on the forward deck, directly over the tank drain and of sufficient size for cleaning the forward compartment and tank drain.

3.5.3 Racks and ladder.

3.5.3.1 Sewage tool rack. A rack shall be provided on the top deck curb side, adjacent to the 125 mm (5-inch) discharge pipe for storage of the four-tine sewer rake, street shovel, scraper, street broom, and agitator hose and tube assembly. The racks shall securely hold the sewage tools from vibrating and coming loose while in transit. Tools shall be of sufficient length to enable the operator to scrape the tank bottom while standing on the tank body ladder or to reach into catch basins to break up so1ids.

3.5.3.2 Hose racks. Sufficient number of racks shall be provided for not less than 4570 mm (15 feet) of 75 mm (3-inch) diameter underground suction hose. The racks shall be accessible from the curb side of the vehicle, and located in such a manner that they can be washed and cleaned. Hose clips shall be provided to hold the hose in the respective racks.

3.5.3.3 Fill hose rack. A rack of sufficient size for 7600 mm (25 feet) of 64 mm (2.5 inch) diameter collapsible fill hose (see 3.5.9.2) shall be provided. The rack shall be accessible from the curb side of the vehicle and located in such a manner that the fill hose may easily be attached to a water hydrant for filling the tank body.

3.5.3.4 Ladder. A ladder shall be securely fastened to the side of the tank body.

3.5.4 Level indicator. A level indicator shall be provided to determine the amount of fluid in the tank and shall be located on the curbside of the vehicle.

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3.5.5 Tank openings.

3.5.5.1 Sewage and waste opening. The tank shall be equipped with a 125 mm (5-inch) diameter pipe which shall be connected to the power hose reel by a flexible coupling for discharging sewage and waste into the rear compartment.

3.5.5.2 Hydrant opening. The tank shall be equipped with a 64 mm (2.5-inch) diameter opening which shall discharge hydrant water into the front compartment of the tank body.

3.5.5.3 Tank drain. A 100 mm (4-inch) tank drain shall be provided to empty the tank body of water directly into a 100 mm (4-inch) diameter pipe to a tee which shall either divert water onto the street or into the centrifugal pump for either agitating sewer basins or flushing. The tank drain shall be located directly below the hatch cover to be accessible for cleaning. The tank drain shall be provided with a 100 mm (4-inch) gate shut-off valve under the tank; quick release rubber coupling to disengage the drain pipe from the tank for hoisting the tank and to prevent vibration; a 75 mm (3-inch) gate shut-off valve and tee connection at the street discharge end. The tee connection shall be piped into the centrifugal pump for adding water under pressure to break solids in catch basins, wash down underground chambers, and flush streets.

3.5.6 Auxiliary engine. An auxiliary engine driven power unit operating on the same type of fuel as the main chassis engine shall be provided.

3.5.6.1 Auxiliary engine mounting and enclosures. The auxiliary engine power unit shall be a liquid cooled, four-stroke cycle, (two or four-stroke cycle for diesel engines) reciprocating, industrial type. The engine shall be mounted on structural steel frame members of sufficient length and strength for the imposed loads to provide mounting of the clutch power takeoff unit and the hydraulic and centrifugal pumps. The entire power unit shall be enclosed and shall be fitted with ventilation panels and hood. The total enclosure shall prevent road splash and rain. Removable access panels, constructed with louvers for ventilation, shall be installed on each side and secured with a spring loaded locking device. Hinges shall be bolted to cabinet doors. The panels and doors shall not vibrate or drum. The panels and doors shall be constructed of 16 gage (1.519 mm) (0.0598 inch) hot rolled mild steel.

3.5.6.2 Engine. The auxiliary engine shall have not less than a 4330 cm^3 (264-cubic inch) displacement and shall provide power for the hydraulic system and the centrifugal pump. The engine shall provide sufficient power to simultaneously operate the centrifugal pump the hydraulic baffle agitator, and the power hose reel.

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3.5.6.3 Engine control devices. Control devices shall be furnished for the engine. Automatic control devices shall include a high temperature control (coolant control set at 104°C (220°F)), an oil pressure control set at the engine manufacturer's minimum recommended oil pressure, and an overspeed control operated independently from the governor mechanism and requiring manual resetting. Manual control devices shall include a choke (gasoline engine only) and a throttle. The manual throttle control shall permit operation of the engine at any selected speed and shall not restrict the action of the governor when the engine in operating at governed speeds.

3.5.6.4 Ignition or starting system. The ignition or starting system shall be a 12-volt electrical circuit with a negative ground. The battery source shall be the truck battery system (see 3.4.2.6).

3.5.6.5 Engine components. The engine fuel supply shall be provided by the truck chassis fuel tank. A fuel pump and filter shall be provided. The muffler shall be positioned so that the fumes do not interfere with the operator.

3.5.6.6 Engine accessories. The following engine accessories shall be panel mounted, identified as to their function, waterproof, and installed in a manner to facilitate removal and service:

(a) Choke (gasoline engine only)
(b) Throttle
(c) Key-operated ignition or starting switch
(d) Oil pressure gage
(e) Tachometer
(f) Starter button
(g) Water temperature gage
(h) Hour meter

The choke (gasoline engine only), throttle, ignition and starter switch shall be in an accessible position to permit the operator to operate the auxiliary engine while standing on the ground.

3.5.7 Power takeoff clutch unit. A power takeoff clutch unit shall be provided, mounted directly on the engine, and controlled by a hand-operated lever located in an accessible position to permit the operator to engage or disengage the clutch while standing on the ground. The power takeoff clutch shall be the manufacturer's standard with torque capacity exceeding maximum delivered engine torque. The power takeoff clutch shall be provided with SAE power takeoff openings. The power takeoff unit shall drive the centrifugal pump.

3.5.8 Centrifugal pump. A high pressure type, single stage centrifugal pump shall be provided. The pump shal1 have a 100 mm (4-inch) suction port and a 75 mm (3-inch) discharge port. The pump shall be capable of pumping water from the tank body and delivering water at

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not less than 690 kilopascals (kPa) (100 pounds per square inch (psi)) pressure to the eductor nozzle and the agitator hose. The pump shall be capable of pumping water directly from underground, through the pump, and into the tank. The centrifugal pump shall be coupled with a flexible coupling to the output shaft of the power takeoff unit.

3.5.8.1 Centrifugal pump components. Centrifugal pump components shall include the following:

(a) Suction and pump cleanout ports
(b) A 6.4 mm (.25-inch) air cock installed on the top of the impeller shell to evacuate air from the suction line.
(c) A 31.4 mm (1.25 inch) drain plug installed on the bottom of the shell to drain the pump
(d) A 25 mm (l-inch) pipe thread opening in the 75 mm (3-inch) pump discharge to locate a 25 mm (l-inch) pipe nipple and shut-off gate valve to which the agitator hose assembly is attached
(e) Provision for attaching the underground suction hose (see 3.5.9.6) to the pump suction intake.

3.5.9 Hoses. All pressure, suction, and discharge hoses shall withstand the intended usage without cracking, collapsing, or swelling and shall have a smooth bore.

3.5.9.1 Tank discharge hose. One 7600 mm (25-foot) length of 64 mm (2.5 inch) discharge hose with quick disconnect cam lock couplings shall be furnished to discharge water from tank body onto street or basin. A means shall be provided while discharging water from tank body or basin while under pressure to prevent hose from jerking.

3.5.9.2 Tank fill hose. One 3530 mm (10-foot) length of 64 mm (2.5 inch) diameter, collapsible fill hose shall be provided and one end shall have a screw-on type coupling compatible with the tank-fill inlet, and the opposite end shall be fitted with an American Standard Taper pipe threaded coupling, so that prior to filling tank, a suitable coupling may be attached for hydrant filling.

3.5.9.3 Suction hose. A suction hose shall be provided and installed as a coupling to prevent vibration between the tank discharge drain tee and the centrifugal pump.

3.5.9.4 Pressure hose. A 75 mm (3-inch) inside diameter (id) pressure hose not less than 5190 mm (17 feet) long shall be provided for connection to the pump discharge and the eductor housing assembly.

3.5.9.5 Sand suction hose. A 100 mm (4-inch) id, smooth bore, wire reinforced sand suction hose not less than 7600 mm (25 feet) in length, shall be furnished. One end of the hose shall be connected to the power hose reel for discharging contents from sewer into rear of tank through

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the hose reel assembly and the 125 mm (5-inch) diameter pipe. The hose shall be connected at the other end to the eductor housing.

3.5.9.6 Underground suction hose. One 4570 mm (15-foot) length of 75 mm (3-inch) id suction hose with quick disconnect type cam lock couplings shall be provided.

3.5.10 Power hose reel assembly. A power hose reel assembly shall be furnished and installed on the front curbside of the tank body, and shall be so constructed that the sand suction hose shall be connected to a pivot. The sewage shall be drawn through the eductor intake opening into the sand suction hose and through the pivot joint, and discharged through a 125 mm (5-inch) diameter pipe extending along the top deck and down into the rear tank compartment. The power hose reel shall be hydraulically controlled and operated by the reel control lever to raise and lower the eductor housing assembly. There shall be a bolted-on cleanout plate located on the pivot joint for cleanout.

3.5.11 Eductor assembly. An eductor assembly shall be furnished and shall remove liquefied sewage from underground. The eductor assembly shall consist of a housing and a nozzle. The eductor housing shall be of cast iron and shall have three openings: a threaded opening for connecting the pressure pipe; an opening for connecting the sand suction hose; and an eductor intake opening. The eductor nozzle shall have a 25 mm (l-inch) diameter orifice, shall be located inside the eductor housing, and shall be replaceable. With the pump delivering water at 620 kPa (90 psi) pressure to the eductor nozzle and with the eductor assembly at ground level, a minimum suction rate of 95 kPa (28 inches) mercury shall be created at the eductor intake opening.

3.5.11.1 Agitator hose and tube assembly. An agitator hose and tube assembly shall be furnished and shall use water under pressure to break up solids in sewer basins enabling the eductor intake opening to draw sewage into the suction hose. The hose and tube assembly shall be rated for a water pressure of 690 kPa (100 psi). A 25 mm (l-inch) id hose, not less than 3050 mm (10 feet) long, shall be furnished and shall be fitted with 25 mm (l-inch), screw-on high pressure fittings. One hose fitting shall connect to the 25 mm (l-inch) fitting on the pressure side of the centrifugal pump (see 3.5.8.1(d)). A 25 mm (l-inch) id tube, not less than 3050 mm (10 feet) long, shall be connected to the other high-pressure hose fitting. The tube shall be fitted on one end with a screw-on type pressure fitting and an adjustable valve.

3.5.11.2 Water suction and pressure connections. All connections and components shall be watertight and air tight to prevent loss of vacuum or pressure and shall withstand the pressure and vacuum required to meet the requirements specified herein.

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3.5.12 Self-propelled sewer nozzle. A self-propelled, pump actuated nozzle for cleaning blocked sewer laterals and culverts shall be provided. Fifteen meters (50 feet) of 64 mm (2.5-inch) diameter sewer cleaning hose for use with the nozzle shall be furnished and stowed on the vehicle.

3.5.13 Hydraulic system. A hydraulic system shall be provided. The hydraulic system shall operate the hydraulic hoist for discharging the tank, shall operate the hose reel for raising and lowering the eductor assembly, and shall operate the baffle agitator for clearing the front baffle of sewage.

* 3.5.13.1 Hydraulic hoist. The hydraulic hoist shall be the manufacturer's standard with a minimum lifting capacity rating of class 50 for size 1200 trucks and class 70 for size 2000 trucks in accordance with the National Truck Equipment Association Dump Hoist Chart. The hoist shall lift the body to a minimum dumping angle of 50 degrees from top of truck chassis frame. Valve and power controls shall be located on the curb side of the vehicle, in a location accessible to an operator standing on the ground.

3.5.13.2 Hydraulic pump. A hydraulic pump shall be provided and shall deliver a constant volume at various speeds. The pump shall have a minimum capacity of 8620 kPa (1250 psi) at 1800 revolutions per minute (rpm) and shall be of sufficient size to provide power to elevate the tank body with full payload, to lower and raise the power hose reel assembly at various depths, and to power the baffle agitator.

3.5.13.2.1 Hydraulic pump oil reservoir. A hydraulic oil reservoir, with a bolted-on cover and a seal with a replaceable filter mounted inside on the discharge port, shall be provided. The oil reservoir shall have a drain plug located underneath the reservoir. The reservoir shall have a splash proof vented fill cap and dip stick with graduation marks. An intake port shall be provided. The reservoir shall have sufficient capacity to operate the hydraulic circuit, plus a 20 percent reserve capacity.

3.5.13.3 Flow control valve. A flow control valve shall be provided and mounted into the system to divert the hydraulic oil either to the hydraulic hoist motor or to operate the hydraulically operated hose reel. If either the hydraulic hoist, hose reel, or baffle agitators are not operating, the hydraulic pump shall continue to circulate oil through the flow control valve and return the oil to the oil reservoir.

3.5.13.4 Three-way valves. Two three-way valves with snubbers shall be provided. One three-way valve shall operate the hose reel motor and hose reel brake cylinder. The second three-way valve shall operate the tank hoist cylinder.

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3.5.13.5 Velocity valves. A secondary hydraulic system shall be provided with velocity valves to prevent the tank body or hose reel from collapsing if a hydraulic line should rupture.

3.5.13.6 Hydraulic hose reel brake system. A hydraulic hose reel brake system shall be furnished and shall actuate upon demand of the operator. The brake shall be of sufficient capacity and size to permit lowering or raising the eductor assembly to any depth that is required. The hydraulic brake shall have built-in devices to prevent any slippage.

3.5.13.7 Hydrau1ic control levers. Control levers for the hydraulically operated devices shall be furnished, shall be accessible to the operator from the curb side of the vehicle while the operator is standing on the ground, and shall be adjacent to the operators work area. The control levers shall be panel mounted, identified as to their function, waterproof, and installed in a manner to facilitate removal and service.

3.5.13.8 Hydraulic pressure lines and fittings. All hydraulic pressure lines shall be steel hydraulic tubing, black pipe, or wire braid hoses with permanently attached fittings. All fittings with threads shall have dry seal fits. Sizes of all hydraulic lines shall conform to the rate of flow and pressure required to operate the hydraulic system smoothly, quietly, and efficiently.

3.5.13.9 Hydrau1ic circuit valve enclosure. Hydraulic valves shall be enclosed with a 16 gage (1.519 mm) (0.0598 inch) steel, waterproof vented enclosure, and shall be installed in a manner to facilitate removal and servicing of the hydraulic circuit.

3.5.14 Cabinet. A cabinet shall be provided on one side of the vehicle. The cabinet shall be of a full-flanged, flush-type construction to carry at least the loads imposed and of a size to house the fresh water tank, sewage tank tools, fittings, and sanitary cleaning supplies. Door(s) shall be top hung and equipped with piano hinges having noncorrosive pins. Hinges shall be bolted to cabinet door(s). Door(s) shall have countersunk locking handles and two point locking mechanism keyed alike. The cabinet shall be constructed of not less than 16 gage (1.516 mm) (0.0598 inch) hot-rolled mild steel. Door(s) shall be constructed to prevent vibration. The cabinet shall be storm proof and shall include provisions for flushing and draining.

3.5.15 Lights. In addition to lighting specified in 3.4.2.3, the following lighting and wiring requirements shall be provided:

(a) A light in the cabinet
(b) Two sealed beam floodlights, mounted in swivel type holders with weatherproof switches, conforming to SAE J598. One floodlight shall be mounted on the left top side at the rear of the truck, and one on the curbside at the forward end of the tank. Both lights shall be accessible from the ground;

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(c) All wiring shall be that which is commercially known as Type No.3, and shall be in conduit with threaded joints, having waterproof and vaporproof connections at the lamps and junction boxes, but shall not include boxes around lights.

3.5.16 Hose reel. A hose reel shall be furnished with 30.5 m (100 feet) of synthetic rubber hose to wash the vehicle and equipment. The hose shall withstand pressures of not less than 1035 kPa (150 psi).

3.5.17 Fresh water tank. A 114 L (30-gallon) freshwater tank shall be furnished and located in the cabinet in an accessible location. The fresh water tank shall provide a reservoir for sanitary cleaning purposes. A method for pressurizing the fresh water tank shall be provided. The tank shall be fastened to cabinet brace reinforcements, and shall have a 25 mm (l-inch) brass American Standard Taper pipe plug with thumb nut for the water inlet. The brass plug shall have a chain with one end attached to the brass pipe plug and the other end fastened to the tank. The tank outlet shall have a T-handle shut-off valve and connection for 13 mm (.50-inch) hose. The tank shall be provided with 7600 mm (25 feet) of 13 mm (.50-inch) id hose for sanitary cleaning purposes. A hose carrier fastened to the inside of cabinet for hose storage shall be provided.

3.5.18 Materials. All valves, eductor housing assembly, clamps, rings, fastening devices, and other materials that are in contact with sewage waste, shall be manufactured from corrosion resistant materials.

3.5.18.1 Dissimilar metals. All dissimilar metals shall be insulated from one another to prevent galvanic action or electrolytic action.

3.5.19 Lubrication system. A lubrication system shall be furnished with fittings for all bearings and other working parts which require relubrication for continued service.

3.5.20 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures, and all rotating or reciprocating parts which are of such nature or so located as to become a hazard to operating personnel, shall be insulated, fully enclosed, or properly guarded.

3.5.21 Instruction plates. Metal plates, showing a schematic layout of all valves, valve controls, water pressure system, auxiliary engine, and equipment with full operating instructions, shall be fastened on the right front corner of the body near the operator's controls.

3.5.22 Drain plugs. Drain plugs shall be provided to completely drain the entire system.

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3.6 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility far inspection. Unless otherwise specified in the contract, contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated

4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected (examined and tested) by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity with the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

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4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and distribution of the curb weight on the front and rear axles. The imposed loading of front and rear axles will be computed using the curb weight, the operator weight at 80 kg (175 pounds), and the payload to provide the specified GVW. Calculated imposed loads on front and rear axles will be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet contract requirements.

* 4.3.2 Road test. The first production vehicle shall be road tested by the contractor for not less than 160 kilometers (km) (100 miles) under the conditions specified in 3.3. Fifty percent of the road test shall be with the vehicle loaded, 50 percent with the vehicle empty. At least 5 sudden stops shall be made from a speed of 32 km/h (20 mph), both loaded and empty.

4.3.3 Pump test. The pump shall be tested to verify conformance to the requirements specified in 3.5.8.

4.3.4 Hydraulic system operational test. The hydraulic system shall be operated to determine that the system functions properly. The test shall include winding and unwinding the power hose reel assembly, hoisting the tank body to the minimum dumping angle specified in 3.5.13.1, and operating the baffle agitator.

4.3.5 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in an as new condition for the duration of the contract.

4.3.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that each vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required in 3.4.25 has been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The vehicle covered by this specification is intended for general administrative use by the Government in cleaning of septic tanks and sewers.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of truck required (see 1.2).
* (c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate service for painting and marking (see 3.1.1.1).
(e) Concealed marking, if required (see 3.1.1.1).
(f) Rustproofing, if required (see 3.1.1.2).
(g) Towing devices on rear, if required (see 3.1.1.4).
(h) A trailer towing package. if required (see 3.1.1.6).
* (i) Operation on JP-4, JP-5 and JP-8 fuels, if required (see 3.4.1.1).
(j) Gasoline engine, if required (see 3.4.1.2).
* (k) Silicone rubber hoses, if required (see 3.4.1.8).
(l) Power plant heaters and fuel warmers, if required (see 3.4.1.9).
* (m) Vernier throttle control, if required (see 3.4.1.10).
* (n) Starting motor circuit breaker, if required (see 3.4.2.1).
(o) Automatic transmission, if required (see 3.4.5.2).
(p) Two-speed axle, if required (see 3.4.9.1).
(q) Traction control, if required (see 3.4.9.3).
(r) Wide base type tires and wheels in lieu of duals, if required (see 3.4.10).
(s) Disc type wheels, if required (see 3.4.10).
(t) Bias ply tires, if required (see 3.4.10.1).
(u) Trailer brake control system, if required (see 3.4.11.2).
(v) Brake controls for use from a towing vehicle, if required (see 3.4.11.3).
(w) Tilt type cab, if required (see 3.4.12).
(x) Individual driver and passenger seats, if required (see 3.4.12).
* (y) Intermittent windshield wipers, if required (see 3.4.14).
* (z) Tools, if required (see 3.4.16.1).
* (aa) Gages in lieu of indicators, if required (see 3.4.19).
* (ab) Engine shutdown system, if required (see 3.4.19).
* (ac) Remote control curbside rearview mirror, if required (see 3.4.20).
* (ad) Heated rearview mirrors, if required (see 3.4.20).
(ae) Air operated horn, in addition, if required (see 3.4.21).
* (af) AM/FM radio, if required (see 3.4.24).
* (ag) Cooling system protection down to -54°C (-65°F), if required (see 3.4.25.1).

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6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number, and vehicle type and class. Unless other conditions are cited in the contract computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.3.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, tabulated 4540 kg (1,000 pounds)) and the engine rpm which is required for 80 km/h (50 mph) geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

* 6.4 Subject term (key word) listing.

Eductor
Nontactical truck
Truck, commercial
Truck, dump type
Truck, sewage and waste
Truck tank
4x2
6x4

6.5 Changes from previous issue. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the last previous issue.

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